Lattice Yang–Mills theory at strong coupling IV: the mass gap with the correct leading asymptotics

A sequel to the foundations monograph of the project "Solving Yang–Mills and the Mass Gap"

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Abstract

For lattice Yang–Mills theory with Wilson's action in dimension $d \geq 3$ and $0 < \beta < \beta_0(d)$ (the threshold of the cluster expansion of [F]), we prove that the mass gap of the infinite-volume theory of Part II and the volume-uniform finite-volume gap of [F] are at least $4\log(1/q(\beta))$, with $q(\beta) = e^3 K(e^{4\beta} - 1)$ and $K = 4(2d-3)$. As $\beta \downarrow 0$ this is $4\log(1/\beta) - O(1)$. The leading coefficient 4 is the one of the strong-coupling expansion (the lightest glueball is a tube with four plaquettes per unit of time, so its mass is $-4\log u(\beta) + o(1)$ with $u(\beta) \sim \beta/4$ for $SU(2)$), whereas the Dobrushin bounds of [F] and Part II give only $\log(1/\beta) - O(1)$. The proof combines three ingredients: duplicate variables, by which every term of the expansion of a covariance that does not connect the two observables cancels by the swap symmetry; the loop expansion and the ratio bound of Part III for the duplicated system; and a geometric lemma: a surface without free edges has at least four temporal plaquettes in each time layer it crosses, because these plaquettes form a graph of minimum degree two on the spatial lattice, whose girth is four. In $d = 2$ the same lemma gives an infinite mass gap in infinite volume and a finite-volume gap growing linearly with the spatial size, in agreement with the exact solution. As in [F], II and III nothing non-abelian is used, so the result belongs to the strong-coupling endpoint of the problem; it is what principle 3 ("the strong-coupling test") of the research program requires of any argument in this regime.

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1 Introduction and main result

The foundations monograph [F] proves that the finite-volume mass gap is bounded below uniformly in the volume for small β, by Dobrushin's method; Part II [II] constructs the infinite-volume theory, identifies its transfer operator with a reversible Markov chain of time slices, and proves the same kind of bound, $m \geq \log(1/(6(d-1)\tanh\beta))$, for the infinite-volume mass gap $m(\mu_\beta)$. These bounds behave like $\log(1/\beta)$ as $\beta \downarrow 0$. The strong-coupling

expansion predicts $m(\beta) = -4\log u(\beta) + O(u)$, where $u(\beta) = c_\rho(\beta)/(Nc_0(\beta))$ ($\approx \beta/4$ for $SU(2)$, $\approx \beta/(2N^2)$ for $N \geq 3$): the lightest state in the gauge-invariant sector propagates as a tube of plaquettes with four plaquettes per unit of Euclidean time. Principle 3 of the research program (`docs/03-research-program.md`, §3.1) asks every argument in the strong-coupling regime to reproduce this behaviour. Here we prove a lower bound with the correct leading coefficient.

Convention 1.1 (Labels). As in [F, Convention 'Labels']: every Theorem, Proposition, Lemma and Corollary below has a complete proof here or in [F, II, III]; Remarks are commentary. We use the notation of [III]: $K := 4(2d-3)$, $\epsilon(\beta) = e^{2\beta} - 1$, $x(\beta) = e^2 K\epsilon(\beta)$, $\theta(\beta) = x/(1-x)$, and $\beta_0(d) = \frac{1}{2}\log(1 + \frac{1}{e^2K(K+2)})$, the threshold below which [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)] holds.

Theorem 1.2 (Main theorem). *Let G, ρ be as in [F, Convention 'Group'] with G metrisable, $d \geq 3$, $0 < \beta < \beta_0(d)$, and*

$$q(\beta) := e^3 K(e^{4\beta} - 1).$$

Then $q(\beta) < \frac{e(2+\epsilon_0)}{K+2} < 0.389$, where $\epsilon_0 = \frac{1}{e^2K(K+2)}$, and:

(i) for every periodic spatial lattice Λ_s with all sides $L_i \geq 4$, the finite-volume mass gap of [F, Definition 'Hamiltonian and mass gap' (`def:H`)] satisfies $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$;

(ii) the mass gap of the infinite-volume theory [II, Definition 'Mass gap of an infinite-volume lattice theory' (`iv:def:gap`)] satisfies $m(\mu_\beta) \geq 4\log(1/q(\beta))$;

(iii) for bounded measurable A, B on $G^{E(\mathbb{Z}^d)}$ depending only on finite link sets S_A, S_B, all of whose endpoints have time coordinates in $[a_1, a_2]$, resp. $[b_1, b_2]$, with $a_2 < b_1$,

$$|\mathrm{Cov}_{\mu_\beta}(A,B)| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1-q(\beta)}\, c(\beta)^{|S_A|+|S_B|}\, q(\beta)^{4(b_1-a_2-2)^+}, \qquad c(\beta) := \Big(e^{2\theta(\beta)}\big(1+\tfrac{1}{K}\big)\Big)^{2(d-1)}.$$

As $\beta \downarrow 0$, $4\log(1/q(\beta)) = 4\log(1/\beta) - 4\log(4e^3K) + O(\beta)$.

Here $\mathrm{Cov}(A,B) := \mu(AB) - \mu(A)\mu(B)$. Statement (iii) is the infinite-volume form of the covariance bound of Theorem 4.1; (i) and (ii) follow from it by [F, Corollary 'Gap from Euclidean correlations' (`cor:gapfromcorr`)] and by [II, Theorem 'OS reconstruction of the lattice theory' (`iv:thm:OS`)](iii).

Remark 1.3 (Comparison). For $d = 4$: the bound of Part II, $\log(1/(18\tanh\beta))$, is $\log(1/\beta) - 2.89 + O(\beta^2)$; the bound here is $4\log(1/\beta) - 29.5 + O(\beta)$. In the common range of validity $\beta < \beta_0(4) \approx 1.54 \cdot 10^{-4}$ the two are comparable near $\beta_0(4)$ (both ≈ 5.6–5.9), and the present one is larger for $\beta \lesssim 1.4 \cdot 10^{-4}$ and by an arbitrarily large factor as $\beta \to 0$ (both computed in `lattice/tests` style with `python3`; see the session journal of 9 November 2026). The point of the theorem is the coefficient 4, not the numerical strength at fixed β; the additive constant is far from the expected $-4\log(1/4)$ for $SU(2)$.

Strategy. (1) For the product measure $\mu\otimes\mu$ of two independent copies (U, U'), $2\,\mathrm{Cov}_\mu(A,B) = \int (A - A')(B - B')\, d(\mu\otimes\mu)$. The duplicated Wilson weight is $\prod_p (1 + f_\beta(U_p))(1 + f_\beta(U'_p)) = \prod_p (1 + g_p)$ with $|g_p| \leq e^{4\beta} - 1$, so the loop expansion of [III] applies to the duplicated system, with partition functions $\Xi(\cdot)^2$. (2) In this expansion every set of plaquettes that does not contain a component connecting the supports of A and B gives a product of two integrals, each odd under the exchange $U \leftrightarrow U'$, hence zero (Lemma 2.5). (3) A connecting component has no free edges away from the supports, so in every time layer between the supports it contains at least four temporal plaquettes (Lemma 3.1), hence at least 4×(number of layers) plaquettes (Corollary 3.3). (4) The sum over such sets is controlled by the ratio bound and the rooted counting of [III].

2 The duplicated system

Throughout this section and the next, $\Lambda = \prod_{\lambda=0}^{d-1} \mathbb{Z}/L_\lambda\mathbb{Z}$ is a periodic lattice with all $L_\lambda \geq 3$, with the conventions of [F, III]: $\mathcal{P} = \mathcal{P}(\Lambda)$, $E(\Lambda)$, $E(p)$, $E(X)$, adjacency and connectedness of sets of plaquettes, $f_\beta = w_\beta/c_0(\beta) - 1$, and for $\mathcal{Q} \subseteq \mathcal{P}$ the restricted partition function $\Xi(\mathcal{Q}) = \int \prod_{p\in\mathcal{Q}}(1 + f_\beta(U_p))\, dU$ of [III, Definition 'Restricted partition functions, loop polymers' (`al:def:Xi`)], so that $\Xi(\mathcal{P}) = Z_\Lambda(\beta)/c_0(\beta)^{|\mathcal{P}|}$.

Definition 2.1 (Duplicated system)**.** On $\Omega_\Lambda \times \Omega_\Lambda$ with the product Haar measure $dU\, dU'$ put, for $p \in \mathcal{P}$,

$$g_p(U, U') := \big(1 + f_\beta(U_p)\big)\big(1 + f_\beta(U'_p)\big) - 1.$$

For $X \subseteq \mathcal{P}$ and bounded measurable F on $\Omega_\Lambda \times \Omega_\Lambda$ let $\Phi_F(X) := \iint F \prod_{p\in X} g_p\, dU\, dU'$.

Lemma 2.2 (Properties of the duplicated system)**.** *(a)* $\|g_p\|_\infty \leq \epsilon(2\beta) = e^{4\beta} - 1$.

(b) For $\mathcal{Q} \subseteq \mathcal{P}$, $\iint \prod_{p\in\mathcal{Q}}(1 + g_p)\, dU\, dU' = \Xi(\mathcal{Q})^2$*; and* $\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}$ *has density* $\prod_{p\in\mathcal{P}}(1 + g_p)/\Xi(\mathcal{P})^2$ *with respect to* $dU\, dU'$.

(c) Let $X \subseteq \mathcal{P}$*, let* ℓ *be a link lying in exactly one plaquette of* X*, and let* H *be bounded measurable and independent of* (U_ℓ, U'_ℓ)*. Then* $\iint H \prod_{p\in X} g_p\, dU\, dU' = 0$.

Proof. (a) By [III, Lemma `al:lem:facts`](b), $1 + f_\beta = w_\beta/c_0 \in [e^{-2\beta}, e^{\beta}]$, so $(1 + f_\beta(U_p))(1 + f_\beta(U'_p)) \in [e^{-4\beta}, e^{2\beta}]$ and $g_p \in [e^{-4\beta} - 1, e^{2\beta} - 1]$; both $1 - e^{-4\beta}$ and $e^{2\beta} - 1$ are at most $e^{4\beta} - 1$. (b) The integrand is $\prod_{p\in\mathcal{Q}}(1 + f_\beta(U_p)) \cdot \prod_{p\in\mathcal{Q}}(1 + f_\beta(U'_p))$; integrate over U and U' separately. For the density: $W_\beta(U)W_\beta(U') = c_0^{2|\mathcal{P}|}\prod_p(1 + g_p)$ and $Z_\Lambda^2 = c_0^{2|\mathcal{P}|}\Xi(\mathcal{P})^2$. (c) Let p be the plaquette of X containing ℓ. Then $U_p = AU_\ell^{\pm 1}B$ and $U'_p = A'U_\ell'^{\pm 1}B'$ with A, B, A', B' independent of (U_ℓ, U'_ℓ), and no other factor of the integrand depends on (U_ℓ, U'_ℓ). By Fubini it suffices that $\iint g_p\, dU_\ell\, dU'_\ell = \int(1 + f_\beta(AU^{\pm 1}B))\, dU \cdot \int(1 + f_\beta(A'U^{\pm 1}B'))\, dU - 1 = 1 \cdot 1 - 1 = 0$, by [III, Lemma `al:lem:facts`](c). □

Definition 2.3 (Loop polymers for a set of links)**.** For $S \subseteq E(\Lambda)$ let $\mathcal{P}_S := \{p : E(p) \cap S \neq \emptyset\}$, let $\mathcal{X}_S$ be the family of sets $X \subseteq \mathcal{P}$ each of whose connected components contains a plaquette of $\mathcal{P}_S$ (including $X = \emptyset$), and for $X \subseteq \mathcal{P}$ let $\overline{X} := \{p : E(p) \cap (E(X) \cup S) \neq \emptyset\}$. Put $J_S := 2(d-1)|S|$. Then $|\mathcal{P}_S| \leq J_S$ and $|\overline{X}| \leq (K+1)|X| + J_S$ (each link lies in $2(d-1)$ plaquettes, and each plaquette is adjacent to at most K plaquettes, [III, Lemma `al:lem:facts`](d)).

Lemma 2.4 (Duplicated loop expansion)**.** *Let* $S \subseteq E(\Lambda)$ *and let* F *be bounded, measurable, and dependent only on the variables* (U_ℓ, U'_ℓ)*,* $\ell \in S$*. Then*

$$\int F\, d(\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}) = \sum_{X\in\mathcal{X}_S} \Phi_F(X) \Big(\frac{\Xi(\mathcal{P} \setminus \overline{X})}{\Xi(\mathcal{P})}\Big)^2.$$

Proof. This is the proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] with f_β replaced by g, single link variables by pairs (U_ℓ, U'_ℓ), dU by $dU\, dU'$, and E_C by S. By Lemma 2.2(b), $\Xi(\mathcal{P})^2 \int F\, d(\mu \otimes \mu) = \sum_{T\subseteq\mathcal{P}} \iint F \prod_{p\in T} g_p\, dU\, dU'$. For $T \subseteq \mathcal{P}$ let $X(T)$ be the union of the components of T meeting $\mathcal{P}_S$ and $Y(T) := T \setminus X(T)$. The proof of the claim in [III, Lemma `al:lem:loopexp`], which uses only the combinatorics of components, of $\mathcal{P}_S$ (there $\mathcal{P}_C$) and of $\overline{X}$, shows that $T \mapsto (X(T), Y(T))$ is a bijection onto the pairs with $X \in \mathcal{X}_S$ and $Y \subseteq \mathcal{P} \setminus \overline{X}$. For such a pair, $F \prod_{p\in X} g_p$ depends only on the variables in the links in $S \cup E(X)$ and $\prod_{p\in Y} g_p$ only on those in $E(Y)$, a disjoint set; so the integral factorises, and $\sum_{Y\subseteq\mathcal{P}\setminus\overline{X}} \iint \prod_{p\in Y} g_p = \iint \prod_{p\in\mathcal{P}\setminus\overline{X}}(1 + g_p) = \Xi(\mathcal{P} \setminus \overline{X})^2$ by Lemma 2.2(b). Divide by $\Xi(\mathcal{P})^2$. □

Lemma 2.5 (Cancellation)**.** *Let A, B be bounded measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B, put $S := S_A \cup S_B$ and $F(U,U') := (A(U)-A(U'))(B(U)-B(U'))$. Then:*

(i) $\int F\, d(\mu\otimes\mu) = 2\,\mathrm{Cov}_\mu(A,B)$ *for every probability measure μ on Ω_Λ, and $\|F\|_\infty \le 4\|A\|_\infty\|B\|_\infty$;*

(ii) if $X \in \mathcal{X}_S$ and no connected component of X contains both a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$, then $\Phi_F(X) = 0$;

(iii) if some link of $E(X)\setminus S$ lies in exactly one plaquette of X, then $\Phi_F(X) = 0$.

Proof. (i) Expand: $\int (A-A')(B-B')\,d(\mu\otimes\mu) = 2\mu(AB) - 2\mu(A)\mu(B)$. (ii) Let X_1 be the union of the components of X that contain a plaquette of $\mathcal{P}_{S_A}$, and $X_2 := X \setminus X_1$; by hypothesis no component of X_1 meets $\mathcal{P}_{S_B}$, and every component of X_2 meets $\mathcal{P}_{S_B}$ but not $\mathcal{P}_{S_A}$. A component containing a link of S_B contains a plaquette of $\mathcal{P}_{S_B}$; hence $E(X_1)\cap S_B = \emptyset$, and likewise $E(X_2)\cap S_A = \emptyset$; and $E(X_1)\cap E(X_2) = \emptyset$, since distinct components share no link. So $S_A \cup E(X_1)$ and $S_B \cup E(X_2)$ are disjoint, and since $dU\,dU'$ is a product measure,

$$\Phi_F(X) = \iint (A-A') \prod_{p\in X_1} g_p \;\cdot\; \iint (B-B') \prod_{p\in X_2} g_p.$$

The exchange $(U,U') \mapsto (U',U)$ preserves $dU\,dU'$ and every g_p (which is symmetric in U_p, U'_p), and changes the sign of $A-A'$; so the first factor equals its negative and vanishes. (iii) Apply Lemma 2.2(c) with $H = F\prod_{p'\in X\setminus\{p\}} g_{p'}$, p the plaquette containing the link. □

3 Geometry: four plaquettes per time layer

Time coordinates are taken in $\mathbb{Z}/L_0\mathbb{Z}$. A link $(x,0)$ is *temporal of layer x_0* (it joins times x_0 and x_0+1); a plaquette $(x,0i)$, $1 \le i \le d-1$, is *temporal of layer x_0*. For $Y \subseteq \mathcal{P}$ and a layer s, Y_s denotes the set of temporal plaquettes of layer s in Y, and $T(p)$ the set of time coordinates of the corners of p (one time for a spatial plaquette, two consecutive times for a temporal one).

Lemma 3.1 (Four plaquettes per layer)**.** *Let $d \ge 3$ and let all spatial sides satisfy $L_i \ge 4$ $(1 \le i \le d-1)$. Let $S \subseteq E(\Lambda)$ and $X \subseteq \mathcal{P}$ be such that every link of $E(X)\setminus S$ lies in at least two plaquettes of X. Let s be a layer none of whose temporal links belongs to S, and let Y be a connected component of X with $Y_s \ne \emptyset$. Then $|Y_s| \ge 4$.*

Proof. The temporal links of layer s are $\ell_{\vec{x}} := ((s,\vec{x}),0)$, $\vec{x} \in \Lambda_s := \prod_{i\ge1}\mathbb{Z}/L_i\mathbb{Z}$, and the plaquettes containing $\ell_{\vec{x}}$ are the temporal plaquettes $((s,\vec{x}),0i)$ and $((s,\vec{x}-e_i),0i)$ of layer s. Let Γ be the graph on Λ_s whose edges are the plaquettes $((s,\vec{x}),0i) \in Y_s$, each joining $\vec{x}$ and $\vec{x}+e_i$ (the positions of its two temporal links). Since $L_i \ge 3$, distinct plaquettes give distinct edges and there are no loops, so Γ is a simple graph. If $\vec{x}$ is an endpoint of an edge, then $\ell_{\vec{x}} \in E(Y)\setminus S$, so $\ell_{\vec{x}}$ lies in at least two plaquettes of X; these are temporal plaquettes of layer s sharing the link $\ell_{\vec{x}}$ with a plaquette of Y, hence they belong to the component Y, i.e. to Y_s. So every non-isolated vertex of Γ has degree at least 2, and since $Y_s \ne \emptyset$, Γ contains a cycle (follow edges without immediately backtracking until a vertex repeats). A cycle in the nearest-neighbour graph of the torus Λ_s has length at least 4 when $d-1 \ge 2$ and all $L_i \ge 4$: a closed walk of length 2 or 3 without backtracking would need $e_i + e_j = 0$ or $e_i \pm e_j \pm e_k = 0$ in Λ_s with unit vectors e_i, e_j, e_k, which is impossible when all $L_i \ge 4$. The edges of the cycle are distinct plaquettes of Y_s. □

Lemma 3.2 (Crossing layers)**.** *Let $Y \subseteq \mathcal{P}$ be connected and let $s \neq s'$ be layers with $Y_s = Y_{s'} = \emptyset$. Then either $T(p) \subseteq I_1$ for all $p \in Y$, or $T(p) \subseteq I_2$ for all $p \in Y$, where $I_1 := \{s+1, \dots, s'\}$ and $I_2 := \{s'+1, \dots, s\}$ are the two cyclic intervals of $\mathbb{Z}/L_0\mathbb{Z}$ determined by s, s'.*

Proof. I_1 and I_2 partition $\mathbb{Z}/L_0\mathbb{Z}$. A spatial plaquette has $T(p) = \{r\}$, contained in one of them. A temporal plaquette of layer $r \notin \{s, s'\}$ has $T(p) = \{r, r+1\}$, and $r, r+1$ lie in the same interval because the only consecutive pairs split between I_1 and I_2 are $\{s, s+1\}$ and $\{s', s'+1\}$. If $p, p' \in Y$ are adjacent, the time coordinates of the endpoints of a common link lie in $T(p) \cap T(p')$, so p, p' lie in the same class. By connectedness all plaquettes of Y lie in the same class. □

Corollary 3.3 (Size of connecting components)**.** *Let $d \geq 3$, all $L_\lambda \geq 3$ and all spatial sides ≥ 4. Let $S_A, S_B \subseteq E(\Lambda)$ and integers $a_1 \leq a_2 < b_1 \leq b_2$ with $b_2 - a_1 < L_0$ be such that the endpoints of the links of S_A have time coordinates in $\{a_1, \dots, a_2\}$ and those of S_B in $\{b_1, \dots, b_2\}$ (modulo L_0). Put*

$$\tau_{\text{in}} := b_1 - a_2 - 2, \qquad \tau_{\text{out}} := L_0 - (b_2 - a_1) - 2.$$

Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus (S_A \cup S_B)$ lies in at least two plaquettes of X, and let Y be a component of X containing a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$. Then $|Y| \geq 4 \min(\tau_{\text{in}}, \tau_{\text{out}})$.

Proof. We may assume $\tau_{\text{in}}, \tau_{\text{out}} \geq 1$. The corners of a plaquette of $\mathcal{P}_{S_A}$ are within time distance 1 of an endpoint of a link of S_A, so their times lie in $W_A := \{a_1 - 1, \dots, a_2 + 1\}$; likewise those of $\mathcal{P}_{S_B}$ lie in $W_B := \{b_1 - 1, \dots, b_2 + 1\}$. Call the layers $s \in \{a_2 + 1, \dots, b_1 - 2\}$ inner (τ_{in} of them) and the layers $s' \in \{b_2 + 1, \dots, a_1 - 2 + L_0\}$ outer (τ_{out} of them). The temporal links of an inner or outer layer join two times of which at least one lies outside $\{a_1, \dots, a_2\}$ and at least one outside $\{b_1, \dots, b_2\}$, so they belong neither to S_A nor to S_B. For an inner s and an outer s', the cyclic intervals of Lemma 3.2 satisfy $W_B \subseteq I_1 = \{s+1, \dots, s'\}$ and $W_A \subseteq I_2 = \{s'+1, \dots, s\}$. If $Y_s = Y_{s'} = \emptyset$ for some inner s and some outer s', Lemma 3.2 puts all of Y in one of the classes I_1, I_2, contradicting the fact that Y contains a plaquette with corner times in W_A and one with corner times in W_B. Hence $Y_s \neq \emptyset$ for all inner layers, or $Y_{s'} \neq \emptyset$ for all outer layers. In either case Lemma 3.1 (with $S = S_A \cup S_B$) gives at least four plaquettes of Y in each of these layers, and plaquettes of different layers are different. □

4 The covariance bound and the mass gap

Theorem 4.1 (Covariance bound in finite volume)**.** *Let $d \geq 3$, $0 < \beta < \beta_0(d)$, and let Λ be a periodic lattice with all $L_\lambda \geq 3$ and all spatial sides ≥ 4. Let A, B be bounded measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B as in Corollary 3.3, with $\tau := \min(\tau_{\text{in}}, \tau_{\text{out}})$. Then, with $q = q(\beta)$ and $c = c(\beta)$ as in Theorem 1.2,*

$$|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A, B)| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1 - q}\, c^{|S_A| + |S_B|}\, q^{4\tau^+}.$$

Proof. Let $S := S_A \cup S_B$ and F as in Lemma 2.5. By Lemmas 2.4 and 2.5, $2\,\mathrm{Cov}(A, B) = \sum_X \Phi_F(X)(\Xi(\mathcal{P} \setminus \overline{X})/\Xi(\mathcal{P}))^2$, where only sets $X \in \mathcal{X}_S$ contribute in which every link of $E(X) \setminus S$ lies in at least two plaquettes of X and some component meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$; by Corollary 3.3 these have $|X| \geq n_0 := 4\tau^+$. For each of them: $|\Phi_F(X)| \leq 4\|A\|_\infty \|B\|_\infty\, \epsilon(2\beta)^{|X|}$ by Lemmas 2.5(i) and 2.2(a); and by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)] and Definition 2.3,

$$\Big(\frac{\Xi(\mathcal{P} \setminus \overline{X})}{\Xi(\mathcal{P})}\Big)^2 \leq e^{2\theta|\overline{X}|} \leq e^{2\theta(K+1)|X|} e^{2\theta J_S} \leq e^{2|X|} e^{2\theta J_S},$$

using $(K+1)\theta < 1$ [III, Lemma `al:lem:constants`]. The number of $X \in \mathcal{X}_S$ with $|X| = n$ is at most $\binom{J_S+Kn}{n}$: this is [III, Corollary 'Counting loop polymers' (`al:cor:countloop`)] with E_C replaced by S (its proof uses only $|\mathcal{P}_S| \le J_S$ and the adjacency bound K). With $\binom{m}{n} t^n \le (1+t)^m$ at $t = 1/K$, $\binom{J_S+Kn}{n} \le (1+\frac{1}{K})^{J_S}(1+\frac{1}{K})^{Kn} K^n \le (1+\frac{1}{K})^{J_S}(eK)^n$. Altogether

$$2|\operatorname{Cov}(A,B)| \le 4\|A\|_\infty \|B\|_\infty \, e^{2\theta J_S}\Big(1+\frac{1}{K}\Big)^{J_S} \sum_{n \ge n_0} \big(eK \cdot e^2 \epsilon(2\beta)\big)^n = \frac{4\|A\|_\infty \|B\|_\infty \, c^{|S|}}{1-q}\, q^{n_0},$$

since $e^3 K \epsilon(2\beta) = q$ and $(e^{2\theta}(1+\frac{1}{K}))^{J_S} = c^{|S|}$. □

Lemma 4.2 (Constants)**.** *For $d \ge 3$ and $0 < \beta < \beta_0(d)$: $q(\beta) < \frac{e(2+\epsilon_0)}{K+2} < 0.389$, and $4\log\frac{1}{q(\beta)} = 4\log\frac{1}{\beta} - 4\log(4e^3K) + O(\beta)$ as $\beta \downarrow 0$. Moreover $\beta_0(d) < \beta_*(d) = \operatorname{artanh}\frac{1}{6(d-1)}$, the threshold of [II].*

Proof. With $\epsilon = e^{2\beta} - 1 < \epsilon_0$ (which is the condition $\beta < \beta_0(d)$), $e^{4\beta} - 1 = \epsilon(2+\epsilon) < \epsilon_0(2+\epsilon_0)$ and $e^3 K \epsilon_0 = \frac{e}{K+2}$; for $d \ge 3$, $K \ge 12$ and $\frac{e(2+\epsilon_0)}{14} < 0.389$. The asymptotics follow from $e^{4\beta} - 1 = 4\beta + O(\beta^2)$. The last claim is [II, Corollary 'Area law in infinite volume' (`iv:cor:arealaw`)] (first line of its proof). □

Proof of Theorem 1.2. The bound on q is Lemma 4.2. (iii) Since $\beta < \beta_*(d)$, the infinite-volume Gibbs measure μ_β of [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)] exists and is the limit of the periodised measures $\tilde{\mu}_{\Lambda_k,\beta}$ on every bounded measurable local function, for every sequence of periodic lattices with all sides tending to infinity. Take such Λ_k with spatial sides ≥ 4. For k large, the reduction map is injective on the links of $S_A \cup S_B$, the functions $A \circ \iota$, $B \circ \iota$ on Ω_{Λ_k} (in the notation of [II]) depend on the images of S_A, S_B, which satisfy the hypotheses of Theorem 4.1 with the same a_i, b_i, $\tau_{\text{in}} = b_1 - a_2 - 2$ and $\tau_{\text{out}} \to \infty$. Theorem 4.1 bounds $\operatorname{Cov}_{\tilde{\mu}_{\Lambda_k,\beta}}(A,B)$ uniformly in k, and the left side converges to $\operatorname{Cov}_{\mu_\beta}(A,B)$.

(ii) Let $F \in \mathcal{A}_+$ in the notation of [II, Definition 'Reflection, shifts, half-space algebras' (`iv:def:reflection`)], depending only on a finite set of links whose endpoints have time coordinates in $[0, n_F]$. Then ΘF depends only on links with endpoint times in $[-n_F, 0]$ and $S_t F$ on links with endpoint times in $[t, t+n_F]$, with the same number of links. For $t \ge 1$ the two sets are disjoint and (iii) gives, by θ- and translation invariance of μ_β (so that $\mu_\beta(\Theta F)\mu_\beta(S_t F) = |\mu_\beta(F)|^2$),

$$\big|\mu_\beta(\Theta F \cdot S_t F) - |\mu_\beta(F)|^2\big| = \big|\operatorname{Cov}_{\mu_\beta}(\Theta F, S_t F)\big| \le C_F \, q^{4(t-2)^+},$$

with C_F independent of t; for $t = 0$ the left side is at most $2\|F\|_\infty^2$. By [II, Theorem `iv:thm:OS`](iii) with $m = 4\log(1/q)$, $m(\mu_\beta) \ge 4\log(1/q)$.

(i) Let A be a bounded gauge-invariant function on $\Omega_s = G^{E(\Lambda_s)}$ and $\Lambda = (\mathbb{Z}/L_0\mathbb{Z}) \times \Lambda_s$ with $L_0 \ge 3$. The functions $\overline{A}(t)$ and $A(0)$ depend on the spatial links of the slices t and 0: in Corollary 3.3 take $a_1 = a_2 = 0$, $b_1 = b_2 = t$; for $1 \le t \le L_0/2$, $\tau = \min(t-2, L_0 - t - 2) = t-2$. Theorem 4.1 gives $|\langle \overline{A}(t) A(0)\rangle - \langle \overline{A}(t)\rangle\langle A(0)\rangle| \le C_A q^{4(t-2)^+}$ with C_A depending on $\|A\|_\infty$ and $|E(\Lambda_s)|$ only, not on L_0. [F, Corollary `cor:gapfromcorr`] (whose proof lets $L_0 \to \infty$ at fixed t, so that the bound is needed only for large L_0) gives $m_{\Lambda_s}(\beta) \ge 4\log(1/q)$. □

5 Two dimensions; discussion

Proposition 5.1 (Two dimensions)**.** *Let $d = 2$ and $0 < \beta < \beta_0(2)$, so that $K = 4$ and $q(\beta) < \frac{e(2+\epsilon_0)}{6} < 0.91$.*

(i) For every periodic lattice with $L_0, L_1 \ge 3$, the spatial lattice $\Lambda_s = \mathbb{Z}/L_1\mathbb{Z}$ has $m_{\Lambda_s}(\beta) \ge L_1 \log(1/q(\beta))$.

(ii) In infinite volume, $\mathrm{Cov}_{\mu_\beta}(A,B)=0$ *whenever the time ranges satisfy* $b_1-a_2\geq 3$*; hence* $m(\mu_\beta)=\infty$*.*

Proof. In $d=2$ the graph Γ of the proof of Lemma 3.1 lives on the cycle $\mathbb{Z}/L_1\mathbb{Z}$ ($L_1\geq 3$, so it is simple), and a subgraph of a cycle in which every non-isolated vertex has degree ≥ 2 is the whole cycle. So the conclusion of Lemma 3.1 becomes $|Y_s|\geq L_1$, that of Corollary 3.3 becomes $|Y|\geq L_1\min(\tau_{\mathrm{in}},\tau_{\mathrm{out}})$, and the proof of Theorem 4.1 gives the bound with $q^{L_1\tau^+}$ in place of $q^{4\tau^+}$; here $q<1$ since $\frac{e(2+\epsilon_0)}{6}<0.91$. (i) then follows as in the proof of Theorem 1.2(i). For (ii), the bound for the periodised observables on tori with $L_1\to\infty$ tends to 0 when $\tau_{\mathrm{in}}\geq 1$, and the torus covariances converge to $\mathrm{Cov}_{\mu_\beta}(A,B)$ as in the proof of Theorem 1.2(iii) (note $\beta_0(2)<\beta_*(2)$); then [II, Theorem `iv:thm:OS`](iii) applies with every $m>0$. □

Remark 5.2 (Consistency with the exact solution). In $d=2$ the transfer matrix on a spatial circle of length L_1 is diagonal in the characters of the holonomy, with eigenvalue ratios $(c_r/(d_rc_0))^{L_1}$ (`lattice/ymlattice/transfer2d.py`, tested against the Monte Carlo); so the exact finite-volume gap is $L_1\log(1/u(\beta))$ for $SU(2)$, linear in L_1, and the infinite-volume theory has no propagating states. Proposition 5.1 reproduces both features (with q in place of u). The difference between $d=2$ and $d\geq 3$ is the girth of the spatial graph (L_1 versus 4): in $d\geq 3$ a closed surface can turn a corner in every time layer with four plaquettes, the minimal glueball.

Remark 5.3 (The strong-coupling test, sharpness, and the $U(1)$ test). (i) Principle 3 of the research program requires the correct asymptotics $m(\beta)\sim 4\log(1/u(\beta))$. Theorem 1.2 gives the lower bound $m\geq 4\log(1/\beta)-O(1)$; the corresponding upper bound would follow from a lower bound on a single correlator (e.g. of two plaquettes), i.e. from showing that the minimal straight tube of $4t$ plaquettes is not cancelled by the other terms of Lemma 2.4. This is expected but not proved here. (ii) The constant $-4\log(4e^3K)$ reflects the crude counting of [III]; counting only surfaces without free edges, as in [III, Remark 'Constants, sharpness, improvements' (`al:rem:constants`)](ii), would improve it. (iii) Nothing here is non-abelian: the argument applies verbatim to $G=U(1)$, whose $d=4$ theory is massless at large β (Part III, Remark on the $U(1)$ test). The result therefore concerns the strong-coupling endpoint only; it shows that the endpoint is now controlled with the right leading behaviour, which is what a renormalisation-group argument would need to reach.

Summary of results

Statement	Where	Status
Duplicated system; loop expansion with Ξ^2; cancellation of non-connecting terms	Lem. 2.2, 2.4, 2.5	Theorem
Surfaces without free edges have ≥ 4 temporal plaquettes per crossed layer ($d\geq 3$)	Lem. 3.1, 3.2, Cor. 3.3	Theorem
Covariance bound $q^{4\tau}$ in finite and infinite volume	Thm. 4.1, 1.2(iii)	Theorem
$m_{\Lambda_s}(\beta)$, $m(\mu_\beta)\geq 4\log(1/q(\beta))=4\log(1/\beta)-O(1)$, $d\geq 3$, $\beta<\beta_0(d)$	Thm. 1.2	Theorem
$d=2$: gap $\geq L_1\log(1/q)$; infinite gap in infinite volume	Prop. 5.1	Theorem
Matching upper bound $m\leq 4\log(1/u(\beta))+O(1)$	Rem. 5.3	open (expected)

References

[F] *Lattice Yang–Mills theory: foundations for the mass gap problem*, Part I of the project "Solving Yang–Mills and the Mass Gap", `theory/lattice-foundations.tex` (this repository), living document.

[II] *Lattice Yang–Mills theory at strong coupling II: the infinite-volume theory, its transfer operator and its mass gap*, `theory/infinite-volume-strong-coupling.tex` (this repository), living document, version of 9 November 2026.

[III] *Lattice Yang–Mills theory at strong coupling III: confinement (area law)*, `theory/area-law-strong-coupling.tex` (this repository), living document, revised 9 November 2026.